📧 Automated Email Notification System

(Built using Automation Anywhere)

🧩 Problem Statement

Organizations often track applications, approvals, or requests using **Google Sheets**.

However, the follow-up process is usually manual:

- ❌ Teams manually check the spreadsheet daily
- ❌ Date calculations are done manually
- ❌ Reminder emails are often delayed or forgotten
- ❌ No proper tracking of "Email Sent" status
- ❌ No centralized error handling

🔎 Resulting Issues

- Communication delays
- Missed follow-ups
- Operational inefficiency
- Human dependency and errors

💡 Automation Idea & Use Case

🚀 Automation Idea

Design and implement an RPA bot in **Automation Anywhere** that:

- Connects to Google Sheets
- Reads application/task data
- Calculates date differences automatically
- Checks status conditions
- Sends reminder emails dynamically
- Updates spreadsheet status
- Handles errors gracefully

🏢 Real-World Use Cases

Industry	Practical Scenario
👩‍💼 HR	Sending follow-ups for pending job applications
💰 Finance	Reminder emails for unpaid invoices
📚 Education	Application status reminders
🛠️ Support	Ticket follow-up notifications
🏢 Admin	Approval reminder automation

👉 Core Concept: **Status-Based + Time-Based Email Trigger Automation**

🔄 Bot Flow Overview (Automation Anywhere Implementation)

1️⃣ Error Handling (Try Block)

- Implemented **Error Handler – Try**
- Ensures safe execution of entire workflow

2️⃣ Connect to Google Sheets

- Google Sheets: Connect
- Open Spreadsheet
- Get Multiple Cells (Read table data)

📌 Retrieves structured data for processing.

3️⃣ Assign Current Date

- `$System:Date$` assigned to `$todayDate$`

📌 Used for calculating time difference.

4️⃣ Loop Through Each Row

- Loop: **For each row in table**

📌 Iterates through all records dynamically.

5️⃣ Convert Sheet Date Format

- Convert `$TableRow[0]` (format: dd/MM/yyyy)
- Store into `$SheetDate$`

📌 Ensures proper datetime comparison.

6️⃣ Calculate Date Difference

- Calculate difference between `$SheetDate$` and `$todayDate$`
- Store result in `$dateDiff$`

📌 Measures number of days elapsed.

7️⃣ Increment Row Index

- Increment `$rowIndex$` by 1

📌 Required to update the correct row in the sheet.

8️⃣ Conditional Logic

Bot checks:

- ✅ `$dateDiff > 2`
- ✅ `$TableRow[4] = "Pending"`

If both conditions are true → trigger action.

9️⃣ Send Reminder Email

- Send email to `$TableRow[2]`
- Subject: **Reminder**

📌 Fully dynamic email automation.

🔟 Update Spreadsheet

- Update cell value to **"Email Sent"**

📌 Prevents duplicate reminder emails.

1️⃣1️⃣ Error Handling (Catch Block)

- Catch All Errors
- Display message: *"Error Encountered"*

📌 Ensures no silent failure.

📊 Final Output / Result

✅ For Records Where:

- Days difference > 2
- Status = "Pending"

The bot:

- 📧 Sends automated reminder email
- 📝 Updates sheet status to "Email Sent"

📌 Example Spreadsheet After Execution

Date	Name	Email	Status	Notification
18/02/2026	Rahul	rahul@email.com	Pending	Email Sent

🎯 Business Impact

- ⚡ 100% automated follow-ups
- ⏱ Zero manual tracking
- 📉 Reduced operational errors
- 📊 Real-time status updates
- 📈 Scalable for large datasets

🧠 Skills Demonstrated

- RPA Development (Automation Anywhere)
- Google Sheets Automation
- Loop & Conditional Logic
- Date-Time Calculations
- Email Automation
- Error Handling (Try–Catch)
- Status-Based Workflow Design

🏁 Conclusion

This project demonstrates a **production-relevant enterprise automation workflow** that eliminates manual tracking and ensures timely communication using structured, scalable RPA design principles.
